

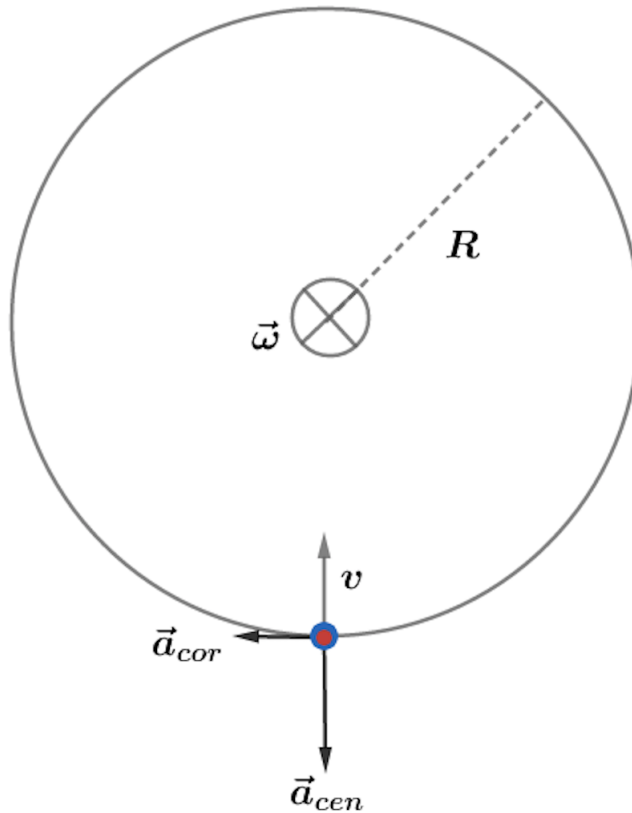
2018A F=ma Exam: Problem 16

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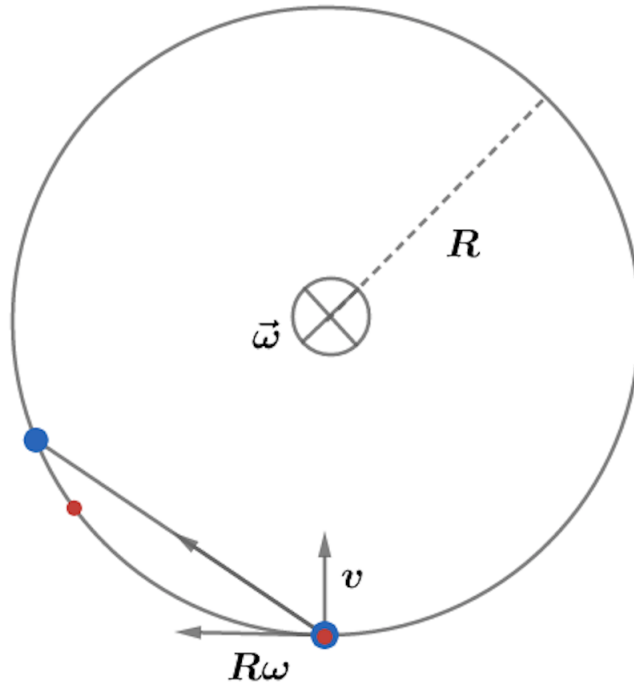
In the noninertial frame of the space station, there are two fictitious forces: centrifugal and Coriolis. They are given by

$$\vec{F}_{cen} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$\vec{F}_{Cor} = -2m\vec{\omega} \times \vec{v}$$



The centrifugal force ensures there is always a component of acceleration directed towards the floor. By following the direction of the Coriolis force, we see the astronaut lands in front of where they started (to the left in the above diagram). This is easier to see in the original inertial frame:



Since the astronaut (blue) travels faster and for a shorter distance, they always land in front of their starting point (red). Thus, the answer is B.